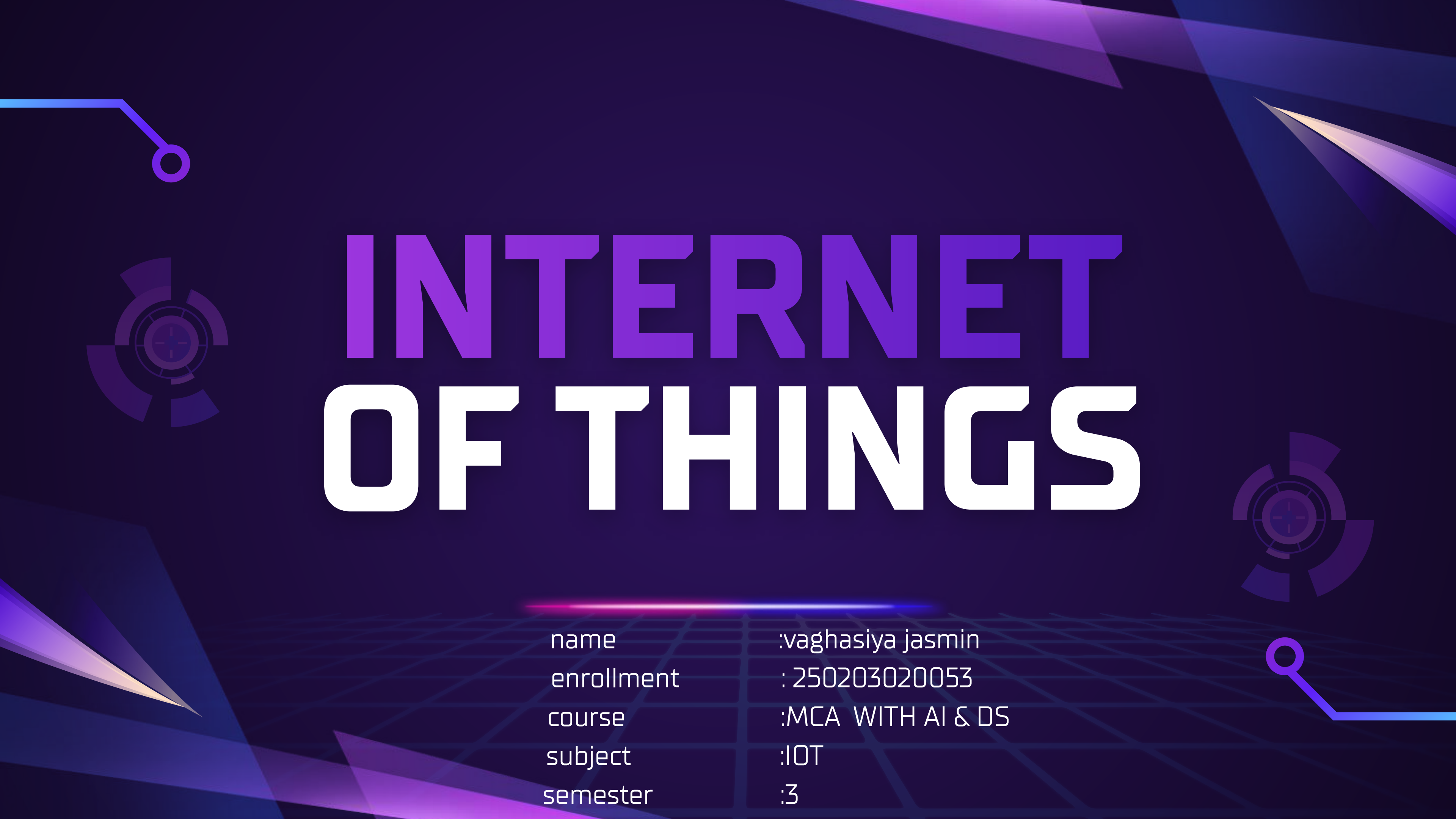
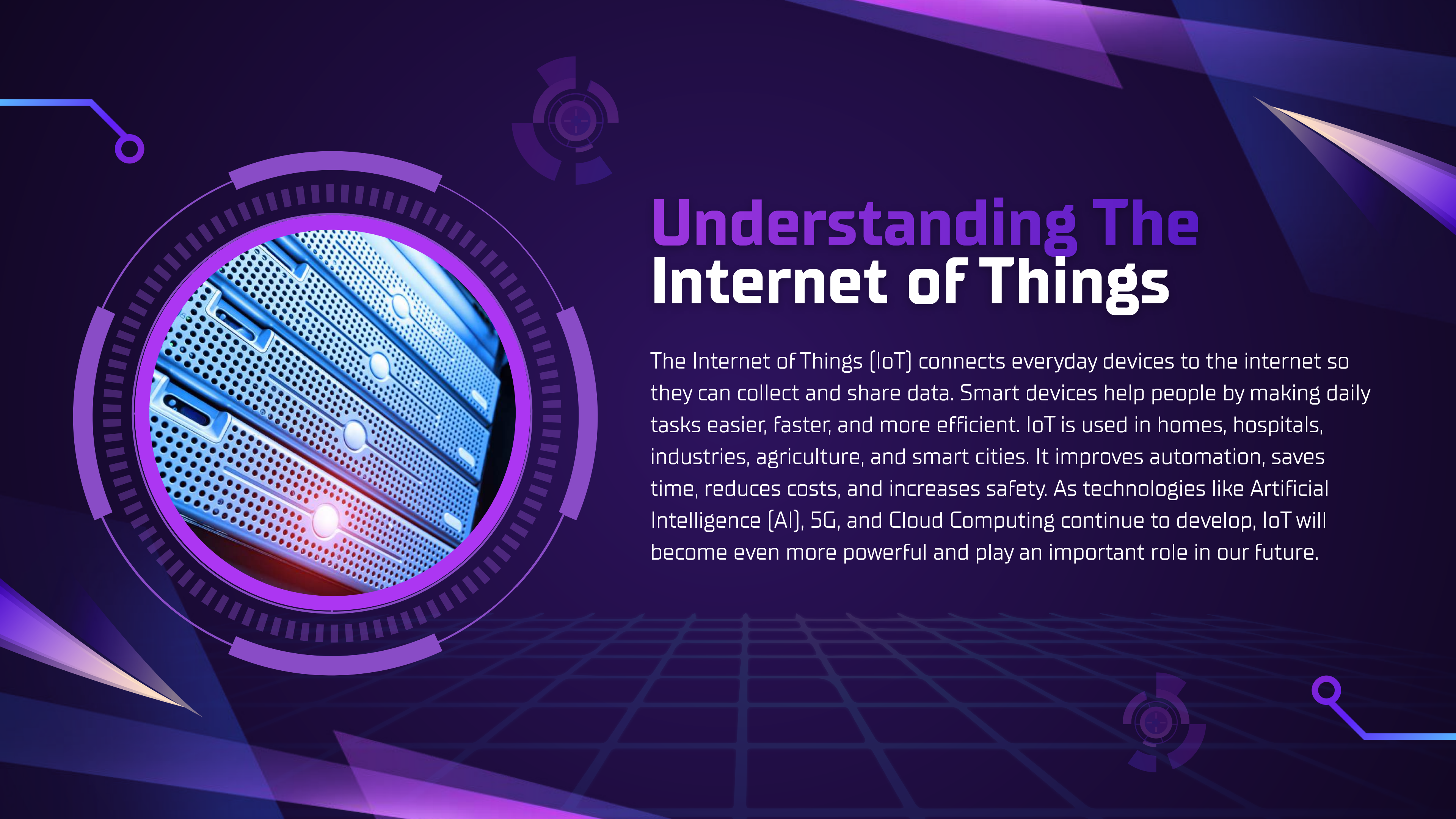




INTERNET OF THINGS

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enrollment	: 250203020053
course	:MCA WITH AI & DS
subject	:IOT
semester	:3



Understanding The Internet of Things

The Internet of Things (IoT) connects everyday devices to the internet so they can collect and share data. Smart devices help people by making daily tasks easier, faster, and more efficient. IoT is used in homes, hospitals, industries, agriculture, and smart cities. It improves automation, saves time, reduces costs, and increases safety. As technologies like Artificial Intelligence (AI), 5G, and Cloud Computing continue to develop, IoT will become even more powerful and play an important role in our future.

TOP IOT COMPANIES



Cisco

Cisco is a global leader in IoT networking, offering end-to-end IoT solutions including secure connectivity, edge computing, and industrial IoT platforms that power smart cities and enterprises worldwide.



IBM

IBM delivers enterprise IoT solutions through its Watson IoT platform, enabling AI-powered analytics, asset management, and predictive maintenance across industries including manufacturing, healthcare, and logistics.



Intel

Intel provides the foundational silicon and hardware technology for IoT devices, offering processors, gateways, and edge computing solutions that power billions of connected devices across industries.



Microsoft

Microsoft Azure IoT Suite offers comprehensive cloud-based IoT services, enabling businesses to connect, monitor, and manage IoT assets at scale with advanced analytics, AI integration, and security.

Future Trends of IoT



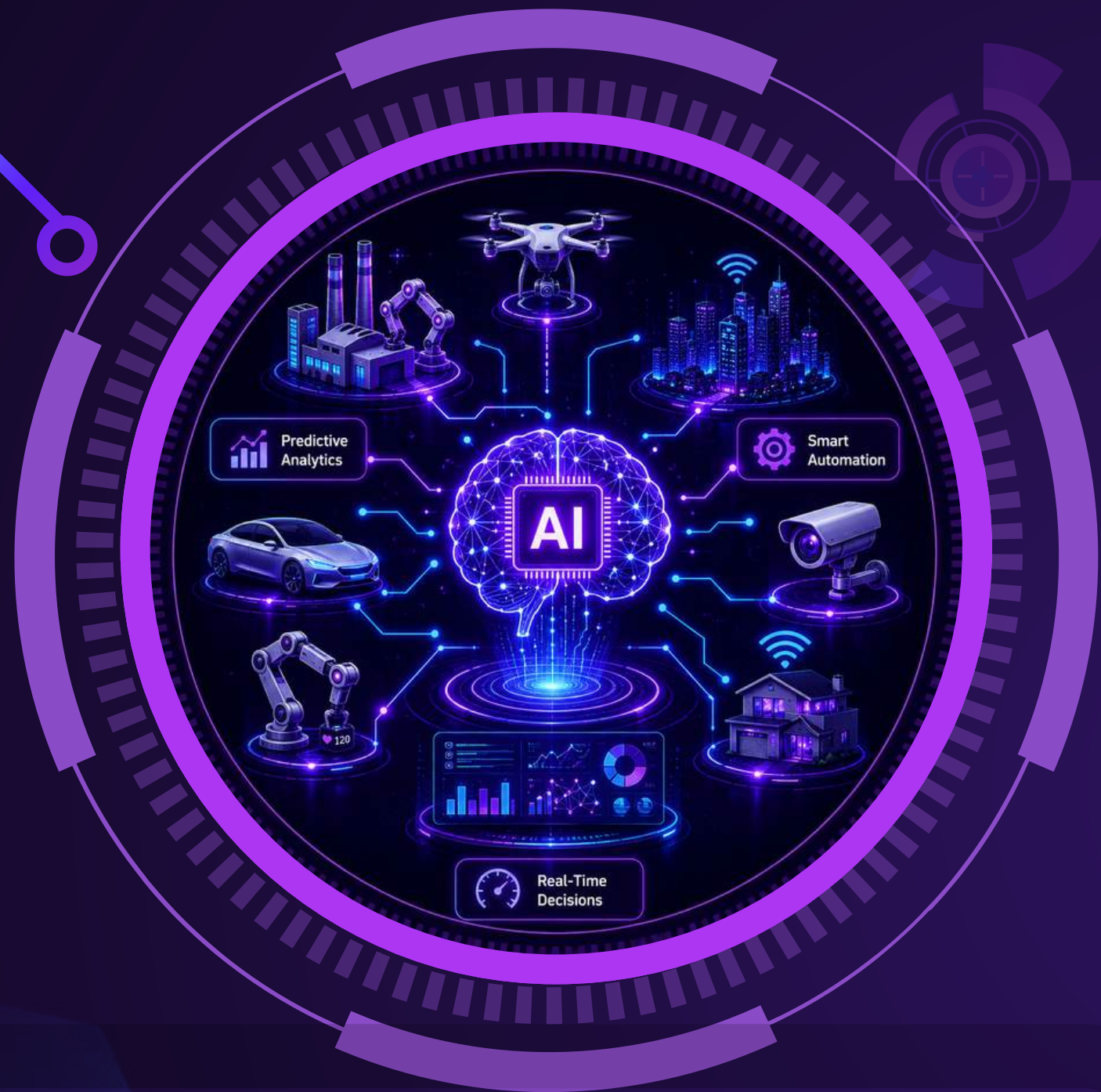
5G Integration

5G provides ultra-fast internet, very low latency, and supports millions of connected devices simultaneously.



Edge Computing

Edge computing processes data near the IoT device instead of sending all data to the cloud.



AI + IoT (AIoT)

5Combines AI with IoT to make devices smarter and capable of automatic decisions.



Smart Cities

Uses IoT for smart traffic, parking, street lighting, waste, and water management.



Healthcare IoT

Enables remote patient monitoring, wearable devices, and smart hospitals.



Industrial IoT

Improves manufacturing with smart factories, predictive maintenance, and automation.



TOP DEVELOPERS IN IoT



1



KEVIN ASHTON



Procter
& Gamble

CONTRIBUTION

- Coined the term "Internet of Things" in 1999.
- Worked at Procter & Gamble.
- Promoted RFID technology.
- Helped develop supply chain automation.

KNOWN AS

Father of IoT Concept



2



VINT CERF



CONTRIBUTION

- Co-invented TCP/IP.
- Called the "Father of the Internet."
- Helped build communication protocols that IoT devices use.

IMPORTANCE

Without the Internet, IoT would not exist.



3



SANJAY SARMA



Massachusetts
Institute of
Technology

CONTRIBUTION

- MIT Professor
- Expert in RFID
- Worked on Auto-ID technology
- Research in supply chain automation

IMPORTANCE

Made RFID practical for IoT applications.



4



ADAM DUNKELS



SWEDISH INSTITUTE OF
COMPUTER SCIENCE

CONTRIBUTION

Created lightweight networking software.

EXAMPLES

- Contiki OS
- uIP
- lwIP

IMPORTANCE

Allows tiny IoT devices with very little memory to connect to the Internet.



5



LIMOR "LADYADA" FRIED



CONTRIBUTION

- Founder of Adafruit Industries
- Promotes open-source hardware
- Provides educational kits for electronics and IoT
- Helps students and developers build IoT projects

IMPORTANCE

Made IoT hardware learning affordable and accessible.



Applications of IoT



1. ENVIRONMENTAL MONITORING

Real-Time Environmental Monitoring using IoT Sensors



BENEFITS



DISASTER PREVENTION

Early detection and real-time alerts help minimize damage and save lives.



POLLUTION CONTROL

Monitors pollution sources and helps maintain a cleaner environment.

Environmental Monitoring

Retail and Logistics



2. RETAIL AND LOGISTICS

APPLICATIONS



SMART INVENTORY
Real-time monitoring of stock levels and automatic updates.



RFID TRACKING
Track products and assets in real-time using RFID tags.



WAREHOUSE AUTOMATION
Automated systems for sorting, picking and packaging.



SHIPMENT TRACKING
Track shipments in real-time and get delivery updates.



BENEFITS



REDUCED THEFT
Real-time tracking and monitoring helps prevent theft and loss.



FASTER DELIVERY
Real-time tracking and route optimization ensure faster and timely deliveries.



BETTER STOCK MANAGEMENT
Accurate inventory data helps avoid overstocking and stockouts.



IoT connects every part of retail and logistics to deliver smarter operations, lower costs and happier customers.



3. SMART TRANSPORTATION

IoT Connected Roads for Smarter, Safer and Sustainable Mobility



APPLICATIONS



SMART TRAFFIC MANAGEMENT

Adaptive traffic signals reduce congestion and improve traffic flow.



PUBLIC TRANSPORT TRACKING

Real-time tracking of buses and trains for better route planning and scheduling.



ACCIDENT DETECTION AND ALERTS

Sensors detect accidents and send instant alerts for quick response.



SMART PARKING

Find available parking spots in real-time and save time and fuel.



EMISSION MONITORING

Monitor vehicle emissions in real-time to reduce pollution and improve air quality.



BENEFITS



IMPROVED SAFETY

Real-time monitoring and alerts help prevent accidents and save lives.



REDUCED CONGESTION

Smart traffic management reduces travel time and traffic jams.



LOWER EMISSIONS

Optimized routes and emission monitoring help create a cleaner environment.



COST EFFICIENCY

Saves fuel, time and operational costs for transport agencies and commuters.



BETTER COMMUTER EXPERIENCE

Real-time information leads to a smoother and more comfortable journey.

Smart Transportation powered by IoT creates intelligent, efficient and eco-friendly mobility for the future.

Smart Transportation



4. SMART HOMES

IoT for Comfort, Convenience and Security



APPLICATIONS



SMART LIGHTING

Control lights remotely and automate based on motion or time.



CLIMATE CONTROL

Smart thermostats optimize temperature for comfort and energy saving.



SMART SECURITY

Smart cameras, door locks and motion sensors ensure 24/7 protection.



SMART APPLIANCES

Control appliances remotely and get real-time status and alerts.



ENERGY MANAGEMENT

Monitor energy usage and optimize consumption to reduce bills.



BENEFITS



ENHANCED SECURITY

Real-time monitoring and smart alerts keep your home safe.



ENERGY EFFICIENCY

Automated control and insights help reduce energy consumption.



COST SAVINGS

Lower utility bills and reduced maintenance costs.



CONVENIENCE

Control your home anytime, anywhere with ease.



BETTER LIFESTYLE

Personalized automation creates a comfortable and smart living.

Smart Homes

HOW IT WORKS



IoT DEVICES



CONNECTIVITY
(Wi-Fi / Zigbee / Bluetooth)



CLOUD PLATFORM



MOBILE APP



SMART AUTOMATION



IoT connects your home, simplifies your life and makes it smarter every day.



5. INDUSTRIAL AUTOMATION

IoT for Smarter, Safer and More Efficient Industries



APPLICATIONS



PREDICTIVE MAINTENANCE

Monitor equipment health in real-time and predict failures before they occur.



REAL-TIME MONITORING

Monitor machines, processes and performance in real-time.



PROCESS AUTOMATION

Automate repetitive tasks and improve production efficiency.



SAFETY & SECURITY

Ensure worker safety and secure industrial operations with smart alerts.



ASSET TRACKING

Track assets and tools in real-time to reduce loss and downtime.

HOW IT WORKS



IoT DEVICES



CONNECTIVITY
(Wi-Fi / 5G / LPWAN)



CLOUD PLATFORM



DATA ANALYTICS
& AI



SMART DECISIONS
& AUTOMATION



BENEFITS



INCREASED PRODUCTIVITY

Optimize operations and improve machine efficiency to boost productivity.



REDUCED DOWNTIME

Predictive maintenance reduces unexpected failures and downtime.



COST SAVINGS

Lower maintenance costs and better resource utilization.



IMPROVED QUALITY

Real-time monitoring ensures consistent quality and reduces defects.



SUSTAINABILITY

Optimize energy usage and reduce waste for a greener industry.



IoT connects machines, people and data to create smart factories of the future.

Industrial Automation

Smart Cities



6. SMART CITIES

IoT for Sustainable, Efficient and Connected Urban Living



APPLICATIONS



SMART LIGHTING

Automatically adjusts brightness and saves energy.



SMART PARKING

Helps drivers find available parking spaces in real-time.



TRAFFIC MANAGEMENT

Monitors traffic flow and reduces congestion using real-time data.



AIR QUALITY MONITORING

Monitors air quality and sends alerts for pollution control.



WASTE MANAGEMENT

Smart bins notify when they are full and optimize collection routes.



WATER MANAGEMENT

Detects leaks, monitors usage and ensures efficient supply.



PUBLIC SAFETY AND SURVEILLANCE

IoT cameras and sensors enhance safety and emergency response.

HOW IT WORKS



IoT DEVICES
(Sensors & Actuators)



CONNECTIVITY
(5G / Wi-Fi / LPWAN)



DATA COLLECTION



DATA ANALYTICS
& AI



SMART DASHBOARD
& ACTIONS



BETTER SERVICES
FOR CITIZENS



BENEFITS



IMPROVED QUALITY OF LIFE

Better services, cleaner environment and safer communities.



ENERGY EFFICIENCY

Reduced energy consumption and lower operational costs.



REDUCED CONGESTION

Real-time traffic control reduces jams and travel time.



COST SAVINGS

Efficient resource utilization reduces overall costs.



SUSTAINABILITY

Optimized water, waste and energy usage for a greener future.



BETTER DECISION MAKING

Data-driven insights help authorities make smart and timely decisions.



IoT connects infrastructure, services and people to build smarter, safer and more sustainable cities.

7. SMART AGRICULTURE

IoT for Smarter Farming, Higher Yields and Sustainable Future

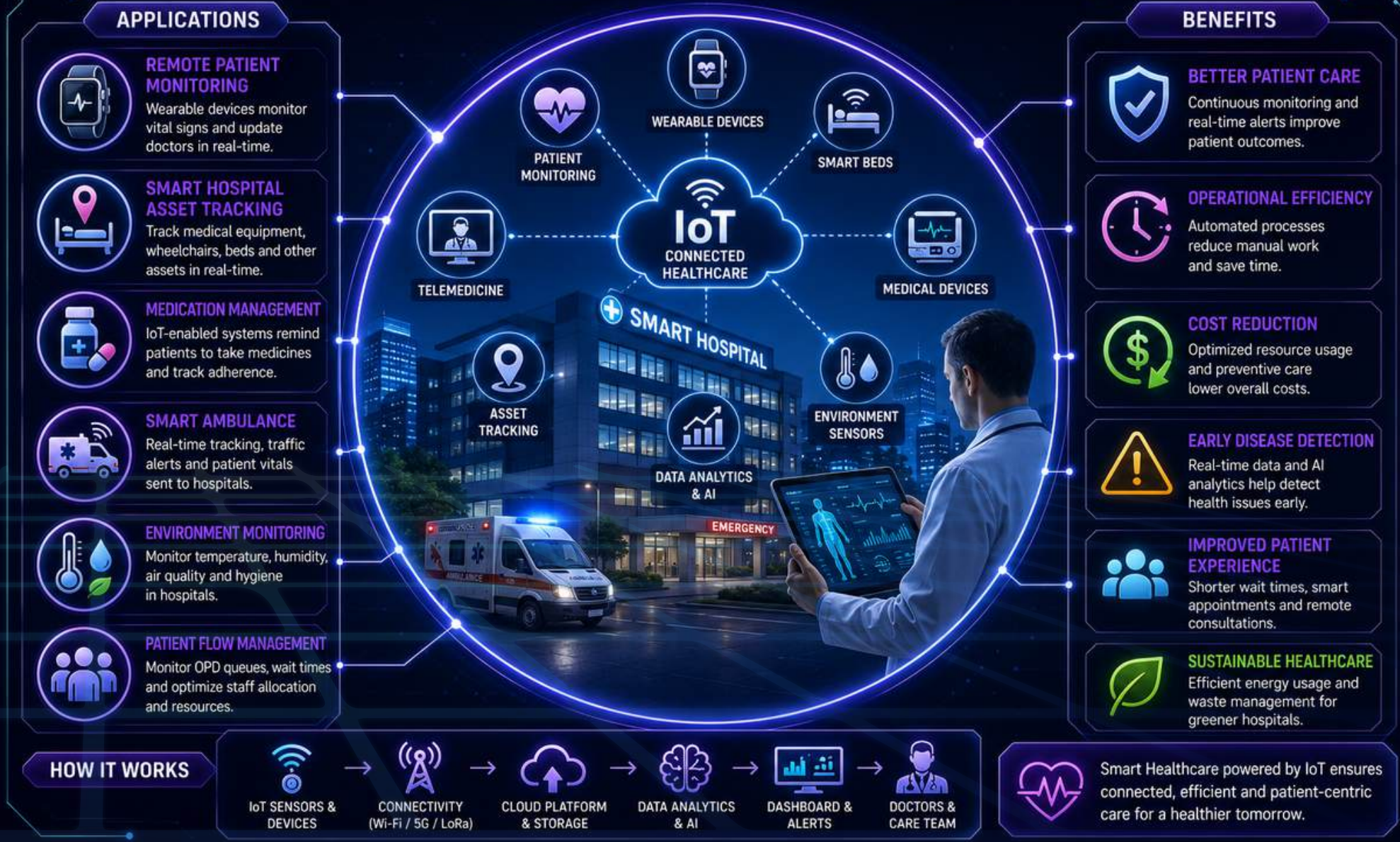


Smart Agriculture

Smart Healthcare

8. SMART HEALTHCARE

IoT for Better Care, Smarter Hospitals and Healthier Lives



Smart Healthcare powered by IoT ensures connected, efficient and patient-centric care for a healthier tomorrow.



THANK YOU
SO MUCH